

General eqn of 2nd degree - Chap-6

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

- i. pair of straight line $\Delta = 0$
- ii. Circle $a = b$ and $h = 0$, $\Delta \neq 0$
- iii. parabola, $\Delta \neq 0$, $ab - h^2 = 0$
- iv. Ellipse, $\Delta \neq 0$, $ab - h^2 > 0$
- v. hyperbola, $\Delta \neq 0$, $ab - h^2 < 0$
- vi. Rectangular hyperbola, $\Delta \neq 0$, $ab - h^2 < 0$ and $a + b = 0$

$a_1 = g\alpha + f\beta + c$ → only for Ellipse and hyperbola

Invariants

$a_1 b_1 = ab - h^2$ → NOT FOR

$a_1 + b_1 = a + b$ → FOR

→ std form

(अब हमें पता है)

Standard form of conics

circle $\rightarrow x^2 + y^2 = a^2$

parabola $\rightarrow y^2 = 4ax$, $x^2 = 4ay$

Ellipse $\rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Hyperbola $\rightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Exercise - (1-10)

Test the nature of the conics

1. $4x^2 + 9y^2 - 8x + 36y - 31 = 0$

compare with,

$$ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$$

$$a = 4, b = 9, g = -4, f = 18, c = -31$$

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$$= (4 \times 9 \times -31) + 2 \times 18 \times -4 \times 0 - 4 \times (18)^2 - 9 \times (-4)^2 - (-31) \times 0$$

$$= -1116 + 0 - 1296 - 144$$

$$\Delta = -2556, \text{ so } \Delta \neq 0$$

Now,

$$\begin{aligned} h^2 - ab &= 0^2 - 4 \times 9 \\ &= -36 < 0 \end{aligned}$$

Now,

$$\begin{aligned} ab - h^2 &= 4 \times 9 - 0^2 \\ &= 36 > 0 \end{aligned}$$

So, It is an ellipse.

$$\textcircled{2} \quad 2x^2 - 3y^2 + 8x + 30y - 27 = 0$$

$$\therefore a = 2, b = -3, h = 0, g = 4, f = 15, c = -27$$

$$\Delta = abc + 2hgf - af^2 - bg^2 - ch^2$$

$$= (2 \times (-3) \times (-27)) + 0 - 2 \times (15)^2 - (-3) \times (4)^2 - 0$$

$$= 162 - 450 + 48$$

$$= -240$$

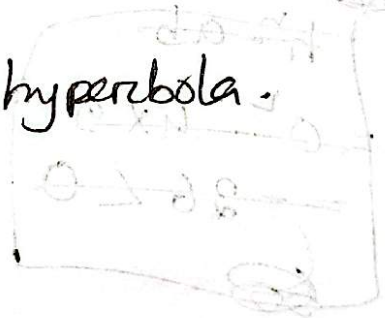
$$\therefore \Delta \neq 0,$$

$$\text{Now, } ab - h^2$$

$$\Rightarrow (2 \times -3) - 0^2$$

$$= -6 < 0$$

So, it is a eqn of hyperbola.



$$\textcircled{3} \quad x^2 - 2y^2 - 6x - 4y + 6 = 0$$

$$a = 1, b = -2, h = 0, g = -3, f = -2, c = 6$$

$$\Delta = 1 \times (-2) \times 6 + 0 - [1 \times (-2)^2] - [(-2) \times (-3)^2] - 0$$

$$= -12 - 4 + 18$$

$$= 2$$

$$\Delta \neq 0, \text{ Now, } ab - h^2$$

$$= 1 \times (-2) - 0^2$$

$$= -2 < 0$$

So, it is an eqn of hyperbola.

$$\textcircled{4} \quad 6x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0$$

$$a = 6, b = -6, h = -5/2, g = 7, f = 5/2, c = 4$$

$$\Delta = [6 \times (-6) \times 4] + 2 \times (5/2) \times 7 \times (-5/2) - 6 \times (5/2)^2 - [(6) \times 7^2] - 4 \times (-5/2)^2$$

$$= -144 - \frac{175}{2} - \frac{75}{2} + 294 - 25$$

$$= 0$$

$$ab - h^2$$

$$= 6 \times (-6) - (-5/2)^2$$

$$= -36 - \frac{25}{4}$$

$$= -\frac{169}{4} < 0$$

$$\textcircled{\circ} \text{ here, } \Delta = 0 \text{ and } ab - h^2 < 0$$

So, ~~it is a~~ it is a pair of straight line

$$(5) (y-a)^2 + (x-a)^2 = 0 \quad (1)$$

$$\Rightarrow y^2 - 2ay + a^2 + x^2 - 2ax + a^2 = 0$$

$$\Rightarrow x^2 + y^2 - 2ax - 2ay + 2a^2 = 0$$

compare,

$$a=1, b=1, g=-a, f=-a, c=2a^2, h=0$$

$$\Delta = 1 \times 1 \times 2a^2 + 0 - 1 \times (-a)^2 - 1 \times (-a)^2 = 0$$

$$= 2a^2 - a^2 - a^2$$

$$= 0$$

~~so, it is an eqn of pair of straight lines~~

$$x=a, y=a$$

$$\text{also, } a=b=1 \text{ and } h=0$$

so, it is a eqn circle.

$$⑥) xy + a^2 = a(x+y)$$

$$\Rightarrow ax + ay - xy - a^2 = 0$$

$$\Rightarrow xy - ax - ay + a^2 = 0$$

$$\text{here, } a=0, b=0, h=\frac{1}{2}, g=-\frac{a}{2}, f=-\frac{a}{2}, c=a^2$$

$$\Delta = 0 + 2 \times \left(-\frac{a}{2}\right) \left(-\frac{a}{2}\right) \left(\frac{1}{2}\right) - 0 - 0 - a^2 \left(\frac{1}{2}\right)^2$$

$$= \frac{a^2}{4} - \frac{a^2}{4}$$

$$= 0$$

$\Delta = 0$, pair of straight lines

$$⑦) x^2 - xy - 2y^2 - x - 4y - 2 = 0$$

$$a=1, h=-\frac{1}{2}, b=-2, g=-\frac{1}{2}, f=-2, c=-2$$

$$\Delta = [1 \times (-2) \times (-2)] + [2 \times \left(-\frac{1}{2}\right) \times \left(-\frac{1}{2}\right) \times (-2)] - [1 \times (-2)^2] - \cancel{(-2) \times 1^2}$$

$$- [(2) \times \left(-\frac{1}{2}\right)^2] - [(-2) \times \left(-\frac{1}{2}\right)^2]$$

$$= 4 - 1 - 4 + \frac{1}{2} + \frac{1}{2}$$

$$= 0$$

$\Delta = 0$, pair of straight lines.

$$\textcircled{8} x^2 + 2xy + y^2 + 2x - 1 = 0$$

$$a=1, b=1, h=1, g=1, f=0, c=-1$$

$$\Delta = (1 \times 1 \times -1) + 0 - 0 - 1 - (-1) \times (1)^2$$

$$= -1 + 0 - 0 - 1 + 1$$

$$= -1$$

$$\Delta \neq 0,$$

$$\Delta \neq 0, ab - h^2$$

$$= 1 \times 1 = 1$$

$$= 0 \rightarrow \boxed{\text{parabola}} \quad \text{Ans}$$

$$\textcircled{9} x^2 + 2xy + y^2 - 6x + 10y + 25 = 0$$

$$a=1, h=1, b=1, g=-3, f=5, c=25$$

$$\Delta = 1 \times 1 \times 25 + [2 \times 1 \times (-3) \times 5] - 1(5)^2 - 1(-3)^2 - 25(1)^2$$

$$= 25 - 30 - 25 - 9 - 25$$

$$= -65 \neq 0, \Delta \neq 0,$$

$$ab - h^2$$

$$= 1 \times 1 - (1)^2$$

$$= 0 \rightarrow \text{parabola}$$

Ans

$$(10) 34x^2 + 24xy + 41y^2 + 48x + 14y - 108 = 0$$

$$a=34, h=12, b=41, g=24, f=7, c=-108$$

$$\Delta = [34 \times 41 \times (-108)] + [2 \times 12 \times 24 \times 7] - [34 \times (7)^2] - [41 \times (24)^2] - [(-108) \times (12)^2]$$

$$= -150552 + 4032 - 1666 - 23616 + 15552$$

$$= -156250, \therefore \Delta \neq 0$$

Now,

$$ab - h^2$$

$$= (34 \times 41) - (12)^2$$

$$= 1394 - 144$$

$$= 1250 > 0$$

→ Ellipse

[Signature]

Reduce the eqn to standard form

Reduce the eqn $32x^2 + 52xy - 7y^2 - 64x - 52y - 148 = 0$ to the standard form.

Soln $32x^2 + 52xy - 7y^2 - 64x - 52y - 148 = 0$ — (1)

Compare,
 $a = 32, b = -7, h = 26, g = -32, f = -26$

$$C = -148$$

$$\Delta = [32 \times (-7) \times (-148)] + [2 \times (-26) \times (-32) \times 26] - [32 \times (-26)^2] - [(-7 \times (-32))^2] - [-148 \times 148 \times 26]$$

$$= 33152 + 43264 - 21632 + 7168 + 100048$$

$$= 162000 \quad \therefore 4 \neq 0(21) - (10 \times 20) =$$

Now, $ab - b^2$

$$= (132x - 7) - (26)^2$$

$$= (132x - 7) - (26)$$

$$= -900 < 0 \rightarrow \text{So it is a eqn of hyperbola.}$$

Contra, $\alpha = \frac{(26x-26) - (-7x-32)}{(32x-7) - (26)^2} \Bigg/ \beta = \frac{(-32x-26) - (32x-26)}{(32x-7) - (26)^2}$

$$\begin{array}{r} -676 - 224 \\ \hline -224 - 676 \end{array}$$

2 1

$$= \frac{0}{-200}$$

≥ 0

Centre (1,0).

Transfer the origin to (1,0) keeping direction of axes unchanged, the eqn (i) become,

$$32x^2 + 52xy - 7y^2 + C_1 = 0 \text{ --- (ii)}$$

Now, we know,

$$C_1 = g\alpha + f\beta + c$$
$$= [-32 \times 1] + (-26 \times 0) - 148$$

$$\Rightarrow C_1 = -180$$

∴ ~~eqn~~ $32x^2 + 52xy - 7y^2 - 180 = 0 \text{ --- (iii)}$

when the xy term is removed by axes, let the reduced eqn will be

$$a_1x^2 + b_1y^2 = 180 \text{ --- (iv)}$$

we know,

$$a_1 + b_1 = a + b$$
$$= 32 - 7$$
$$= 25$$

and

$$a_1b_1 = ab - h^2$$
$$= (32 \times -7) - (26)^2$$
$$= -224 - 676$$
$$= -900$$

$$\therefore (a_1 - b_1)^2 = (a_1 + b_1)^2 - 4a_1b_1$$
$$= 625 - 4(-900)$$

$$(a_1 - b_1)^2 = 4225$$

$$\therefore a_1 - b_1 = 65$$

Solving $a_1 + b_1 = 25$

and $a_1 - b_1 = 65$

We get, $a_1 = 45, b_1 = -20$

from eqn (IV)

$$45x^2 - 20y^2 = 180$$

$$\Rightarrow \frac{x^2}{\frac{1}{4}} - \frac{y^2}{\frac{9}{5}} = 1$$

↳ it is the reduce standard form of the eqn.

14(c) $x^2 + 12xy - 4y^2 - 6x + 4y + 9 = 0$, reduce the eqn to the std form and identify it.

Solⁿ: $x^2 + 12xy - 4y^2 - 6x + 4y + 9 = 0$ — (1)

$$a=1, b=-4, h=6, g=-3, f=2, c=9$$

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$$= (1 \times -4 \times 9) + [2 \times 2 \times -3 \times 6] - 1(2)^2 - (-4) \times (-3)^2 - 9 \times (6)^2$$

$$= -36 - 72 - 4 + 36 - 324$$

$$= -400 \neq 0$$

$$\therefore ab - h^2 =$$

$$= 1 \times -4 - (6)^2$$

$= -40 < 0$; the given conic represents a hyperbola.

Center (α, β)

$$\alpha = \frac{hf - bg}{ab - h^2}$$

$$= \frac{6 \cdot 2 + (-4) \times (-3)}{1 \times (-4) - 6^2}$$

$$= \frac{12 - 12}{-4 - 36}$$

$$= 0$$

$$\beta = \frac{hg - af}{ab - h^2}$$

$$= \frac{(6 \cdot -3) - (1 \cdot 2)}{1 \times (-4) - 6^2}$$

$$= \frac{-18 - 2}{-40}$$

$$= \frac{1}{2}$$

$\therefore$ center $(0, \frac{1}{2})$

Transfer the origin to $(0, \frac{1}{2})$ keeping direction of axes unchanged, the eqn (1) become

$$x^2 + 12xy - 4y^2 + c_1 = 0$$

we know,

$$c_1 = g\alpha + f\beta + c$$

$$= (-9 \times 0) + (2 \times \frac{1}{2}) + 9$$

$$= -10$$

$$\therefore x^2 + 12xy - 4y^2 + 10 = 0 \quad \text{--- (11)}$$

Now, ~~transform~~ when the xy -term is removed by axes, let the reduce eqn will be

$$a_1x^2 + b_1y^2 = -10 \quad \text{--- (11)}$$

we know,

$$a_1 + b_1 = a + b$$

$$= 1 - 4$$

$$= -3$$

$$a_1b_1 = ab - h^2$$

$$= -4 - (6)^2$$

$$= -40$$

$$(a_1 - b_1)^2 = (a_1 + b_1)^2 - 4a_1b_1$$

$$= (-3)^2 - 4 \times (-40)$$

$$= 9 + 160$$

$$(a_1 - b_1)^2 = \cancel{169} + 160$$

$$\therefore a_1 + b_1 = 13$$

Solving $a_1 + b_1 = -3$

$$a_1 - b_1 = 13$$

We get, $a_1 = 5, b_1 = -8$

$$\therefore 5x^2 - 8y^2 = -10$$

$$\Rightarrow \frac{x^2}{-2} + \frac{y^2}{\frac{5}{4}} = 1 \quad \rightarrow \text{which is required std form.}$$

14(e) Find the nature of conic represented by $5x^2 + 2xy + 5y^2 + 10x - 22y + 29 = 0$ convert it into the std form.

Soln $5x^2 + 2xy + 5y^2 + 10x - 22y + 29 = 0 \quad \text{--- (1)}$

$$a=5, b=5, h=1, g=5, f=-11, c=29$$

$$\begin{aligned} \Delta &= abc + 2fgh - af^2 - bg^2 - ch^2 \\ &= 5 \times 5 \times 29 + 2 \times -11 \times 5 \times 1 - 5(-11)^2 - 5(5)^2 - 29(1)^2 \\ &= 725 - 110 - 605 - 125 - 29 \\ &= -144 \neq 0 \end{aligned}$$

$$\begin{aligned} &ab - h^2 \\ &= 5 \times 5 - 1^2 \end{aligned}$$

$= 24 > 0 \rightarrow$ The given conic represented an ellipse.

Let, centre (α, β)

$$\alpha = \frac{hf - bg}{ab - h^2}$$

$$= \frac{(1 \times -11) - 5 \times 5}{(5 \times 5) - (1)^2}$$

$$= \frac{-11 - 25}{24}$$

$$= -\frac{3}{2}$$

$$\beta = \frac{hg - af}{ab - h^2}$$

$$= \frac{(1 \times 5) - (5 \times -11)}{(5 \times 5) - (1)^2}$$

$$= \frac{5 + 55}{24}$$

$$= \frac{5}{2}$$

centre $(-\frac{3}{2}, \frac{5}{2})$

Transfer the origin to $(-\frac{3}{2}, \frac{5}{2})$ keeping direction of axes unchanged, the eqn ① become,

$$5x^2 + 2xy + 5y^2 + C_1 = 0$$

$$C_1 = xg + f\beta + c$$

$$= 5x \cdot \frac{3}{2} + \frac{5}{2} \times 2x - 11 + 29$$

$$= -6$$

$$\Rightarrow 5x^2 + 2xy + 5y^2 - 6 = 0 \text{ --- ②}$$

Now, when the xy term is removed by axes,

let the reduce eqn will be

$$a_1x^2 + b_1y^2 = 6 \quad \text{--- (iii)}$$

$$\begin{array}{l|l} a_1 + b_1 = a + b & a_1 b_1 = ab - h^2 \\ = 5 + 5 & = 5 \times 5 - 1^2 \\ = 10 & = 24 \end{array}$$

$$(a_1 - b_1)^2 = 10^2 - 4 \times 24$$

$$= 100 - 96$$

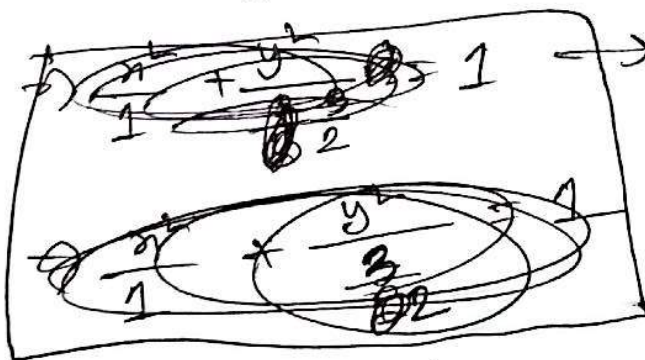
$$(a_1 - b_1)^2 = 4$$

$$\therefore a_1 - b_1 = 2$$

Solving $a_1 + b_1 = 10$, $a_1 - b_1 = 2$

$$\therefore a_1 = 6, b_1 = 4$$

$$6x^2 + 4y^2 = 6$$



which is the reduce std eqn.

$$\Rightarrow \frac{x^2}{1} + \frac{y^2}{\frac{3}{2}} = 1$$